



Computer Concepts & Characteristics of Computers

Grade 10 · ICT · Module 2.1

ICT Grade 10 - Chapter 2 - Module 1 - Akila Savinda

LEARNING OBJECTIVES

What You Will Learn

01

Define a computer

Understand what a computer is and how it works

02

Identify key characteristics

Explore the five main features of a computer

03

Classify by size

Super, Mainframe, Mini, and Micro computers

04

Classify by technology

Analog, Digital, and Hybrid computers

05

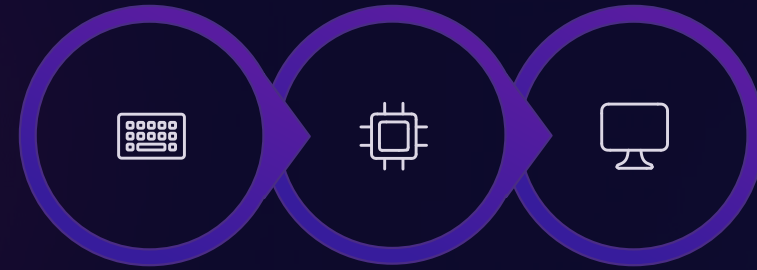
Compare computer types

Understand the differences between each category



What is a Computer?

"A computer is an electronic device that accepts or collects data, processes it according to given instructions, and produces the desired output."



Input

Process

Output

Computers have become an **essential device** in modern life – from education and healthcare to business and entertainment. Recall from Lesson 01: **Data → Processing → Information**.

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Features of a Computer: Speed & Accuracy

⚡ Speed

A computer executes **more than one million instructions per second**. Complex calculations that would take humans days can be completed in milliseconds.

🎯 Accuracy

A computer provides **correct output when correct data is given**. It does not make mistakes on its own — but wrong input leads to wrong results.



GIGO: Garbage In, Garbage Out — the quality of output depends entirely on the quality of input.

Features of a Computer: Efficiency & Versatility



Efficiency

A computer **never gets tired**. It can work round the clock with the same level of accuracy. Banking systems, hospital monitors, and online services run 24/7 without fatigue.

Versatility

A computer can **perform many different tasks simultaneously** – preparing documents, controlling machinery, diagnosing illnesses, playing games, and designing buildings.



Feature 5: Storing & Retrieving

A computer can **store large amounts of data in a small space** and retrieve it quickly when needed.



E-Readers

Thousands of books stored on one small device



Smartphones

Entire music libraries in your pocket



Hospitals

Patient records retrieved instantly when needed



Discussion: Which computer feature do you think is most important, and why?

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Classification of Computers

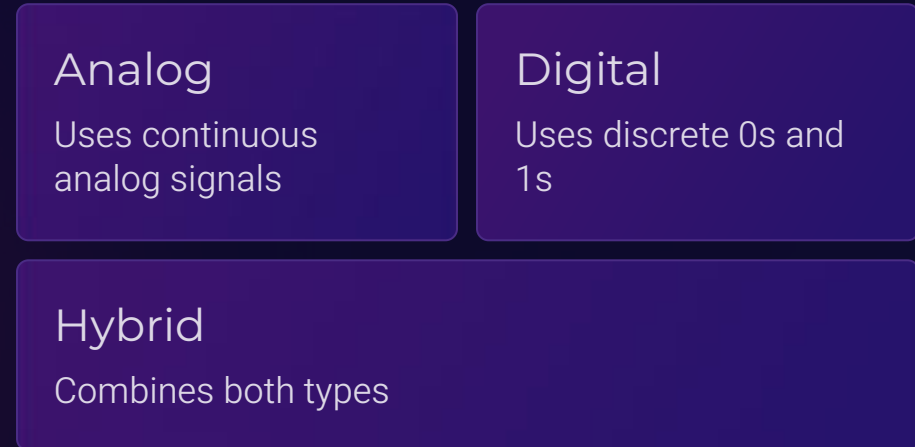
Computers can be classified in two main ways – by their **size** and by their **technology**.

By Size



- 1 Super Computers
- 2 Mainframe Computers
- 3 Mini Computers
- 4 Micro / Personal Computers

By Technology



Classification by Size: Super & Mainframe



SUPER COMPUTERS

The **most powerful** computers — large, expensive, and rare. Used by NASA, military, and for complex scientific calculations.

- Highest computing power available
- Solve complex mathematical problems
- Very large and very expensive

MAINFRAME COMPUTERS

Less powerful than supercomputers but support **many users via terminals**. Common in large businesses and e-commerce.

- Multiple users share one system
- Expensive, but less than supercomputers
- Used for large-scale data processing

Mini & Personal Computers



Mini Computers

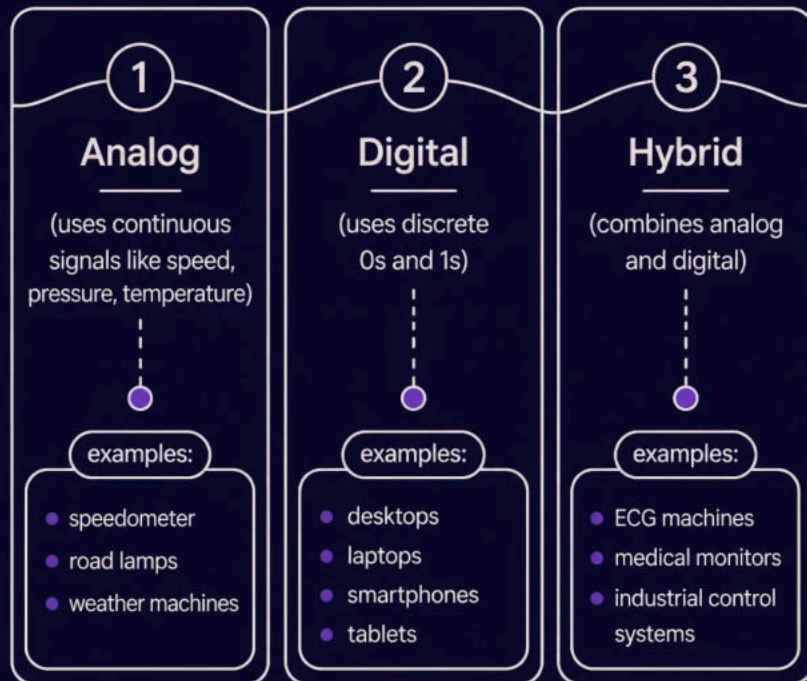
Slower and less expensive than mainframes. A **few users** connect via terminals. Common in some banking systems and small institutions.



Personal Computers (Micro)

The **most common type** — designed for individual use. Affordable, small, and versatile. Includes desktops, laptops, tablets, and smartphones.

Classification by Technology



How Hybrid Computers Work

Hybrid computers **combine analog and digital** technology. For example, an ECG machine:

1. Identifies the heartbeat (analog signal)
2. Converts it to a digital signal
3. Processes and prints the output



Activity: Find other examples of hybrid computers in daily life.